The No-U-Turn Sampler

Adaptively Setting Path Lengths in Hamiltonian Monte Carlo

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MCMC & Hamiltonian Monte Carlo

- Exact posterior inference is often intractable; MCMC is asymptotically unbiased [Nea93].
- Random-walk & Gibbs [Met+53; GG84] struggle in high-dimensional or correlated spaces.

Hamiltonian Monte Carlo

Auxiliary momentum $r \sim \mathcal{N}(0, I) +$ leapfrog: $p(\theta, r) \propto \exp\{\mathcal{L}(\theta) - \frac{1}{2} r^\top r\}$. Volume-preserving $\Rightarrow$ proposals can be **distant** yet **highly probable**.

Good L requires costly preliminary runs and expert knowledge, a key barrier to adoption.

Step size ε

Too small $\Rightarrow$ wasted computation

Too large $\Rightarrow$ high rejection rate

Leapfrog steps L

Too small $\Rightarrow$ random walk

Too large $\Rightarrow$ trajectory loops back

NUTS addresses both tuning problems simultaneously.

NUTS Methodology

No-U-Turn criterion

Stop when the trajectory doubles back: $(\theta^+ - \theta^-)\cdot r^- < 0$ or $(\theta^+ - \theta^-)\cdot r^+ < 0$.

Recursive doubling

1. Draw direction $v_j \sim \mathrm{Unif}(\{-1, +1\})$
2. Take 2^j leapfrog steps of size $v_j \varepsilon$
3. Grow balanced binary tree; halt on U-turn

Memory $\mathcal{O}(2^j) \to \mathcal{O}(j)$: partition $\mathcal{C} = \mathcal{C}^{\mathrm{old}} \cup \mathcal{C}^{\mathrm{new}}$, propose $w' \in \mathcal{C}^{\mathrm{new}}$, accept with $\min\left(1, \frac{|\mathcal{C}^{\mathrm{new}}|}{|\mathcal{C}^{\mathrm{old}}|}\right)$.

Dual averaging (warm-up only)

Drive acceptance to target δ:

$$x_{t+1} = \mu - \frac{\sqrt{t}}{\gamma(t + t_0)} \sum_{i=1}^{t} H_i,$$

$$\bar{x}_{t+1} = t^{-\kappa} x_{t+1} + (1 - t^{-\kappa}) \bar{x}_t$$

$x = \log \varepsilon$, $\mu = \log(10\varepsilon_0)$

Fixed Hyperparams: $\gamma = 0.05$, $t_0 = 10$, $\kappa = 0.75$

No-U-Turn criterion + dual averaging $\Rightarrow$ NUTS is **entirely tuning-free**.

Evaluating NUTS: The Paper's Benchmarks

The paper tests NUTS on four **unimodal or mildly structured** targets.

250-D	25-D	302-D	3001-D
Multivariate Normal	Logistic Regression	Hierarchical LR	Stochastic Volatility

NUTS matches or exceeds best-tuned HMC on **all four benchmarks** without any trajectory-length tuning. What about posteriors with complex geometry?

Multimodality

BNN, 191-D

$5! \times 5! =$

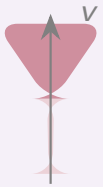
14400 modes

Dense Correlations

GP Classif., 150-D

dense 150×150 cov.

Varying Geometry



Neal's Funnel, 10-D

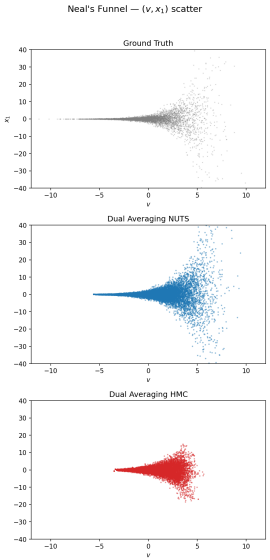
spatially varying σ

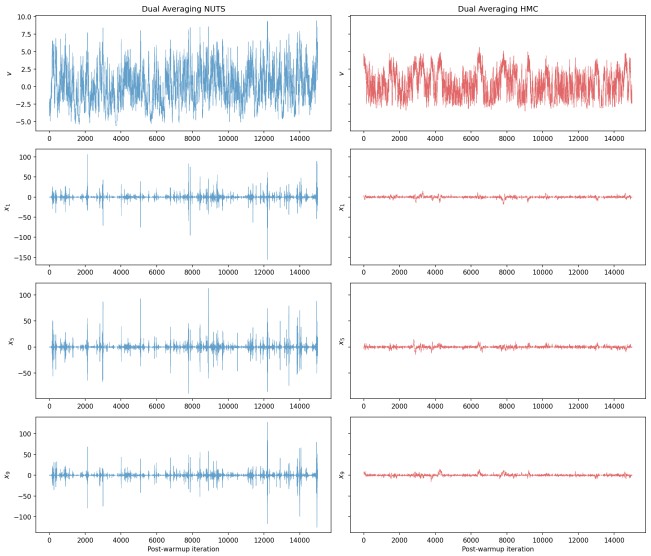
Beyond the Paper: Three Stress Tests

Results

- **BNN:** NUTS explores wider weight space but lower ESS/grad than well-tuned HMC (0.03 vs. 1.57×10^{-4}).
- **GPC:** Modest 40% ESS/grad gain, but NUTS uses $4\times$ more gradient evaluations; advantage is narrow.
- **Funnel:** NUTS recovers $\mathrm{Var}(v) \approx 9$ (truth); HMC's fixed ε fails to explore the neck.

Adaptive trajectory length is critical on complex posteriors.





Summary & Contributions

Paper assessment

+ Resolves HMC's central practical limitation; conceptually simple yet theoretically rigorous
+ No-U-Turn criterion + recursive doubling preserve detailed balance
+ Dual averaging for step-size adaptation $\Rightarrow$ essentially tuning-free
− Dual-averaging hyperparameters justified empirically; robustness across geometries is open
− Performance guarantees mainly asymptotic; limited range of posterior geometries tested

Our contributions

- **Proofs** of uniform $\mathcal{C}$ distribution and detailed balance of the memory-efficient kernel (only sketched in paper)
- **Reproduced** all 4 benchmarks; NUTS matches or exceeds best-tuned HMC without trajectory-length tuning
- **Stress tests:** adaptive trajectory length is critical on complex posteriors, the gains are most dramatic on Funnel & BNN; advantage narrows on dense GP
- **Conclusion:** confirms NUTS's suitability as a general-purpose inference engine for modern Bayesian models

Bibliography i

[Met+53] Nicholas Metropolis et al. **"Equation of state calculations by fast computing machines".** In: *The journal of chemical physics* 21.6 (1953), pp. 1087–1092.

[GG84] Stuart Geman and Donald Geman. **"Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images".** In: *IEEE Transactions on pattern analysis and machine intelligence* 6 (1984), pp. 721–741.

[Nea93] Radford M Neal. **"Probabilistic inference using Markov chain Monte Carlo methods".** In: (1993).

Appendix: The Transition Kernel & Detailed Balance

To reduce memory from $\mathcal{O}(2^j)$ to $\mathcal{O}(j)$, NUTS partitions the candidate set: $\mathcal{C} = \mathcal{C}^{\text{old}} \cup \mathcal{C}^{\text{new}}$.

The Proposal:

- From state $w \in \mathcal{C}^{\text{old}}$, propose a jump to $w' \in \mathcal{C}^{\text{new}}$ uniformly.
- Accept with probability: $\min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right)$

Proof of Detailed Balance (Intuition)

We must satisfy: $p(w|\mathcal{C})T(w'|w,\mathcal{C}) = p(w'|\mathcal{C})T(w|w',\mathcal{C})$

Substituting the prior $p(w|\mathcal{C}) = \frac{1}{|\mathcal{C}|}$ and the acceptance rates:

$$\frac{1}{|\mathcal{C}||\mathcal{C}^{\text{new}}|} \min\left(1, \frac{|\mathcal{C}^{\text{new}}|}{|\mathcal{C}^{\text{old}}|}\right) = \frac{1}{|\mathcal{C}||\mathcal{C}^{\text{old}}|} \min\left(1, \frac{|\mathcal{C}^{\text{old}}|}{|\mathcal{C}^{\text{new}}|}\right)$$

Proving the kernel preserves the target distribution.

Appendix: Uniform Distribution over the Candidate Set

Claim: $p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) = \frac{1}{|\mathcal{C}|}\, \mathbb{I}[w \in \mathcal{C}]$

Let $w = (\theta, r)$. The augmented joint density is
$p(\theta, r, u) \propto \mathbb{I}\big[u \leq \exp\big(\mathcal{L}(\theta) - \frac{1}{2} r \cdot r\big)\big]$.

Conditioning on $(u, \mathcal{B}, \mathcal{C}, \varepsilon)$:

$$p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) \propto \underbrace{p(\mathcal{B}, \mathcal{C} \mid w, u, \varepsilon)}_{\text{(ii)}} \cdot \underbrace{p(w \mid u)}_{\text{(i)}}$$

(i) C.3: $w \in \mathcal{C}$ implies the slice constraint holds $\Rightarrow p(w \mid u) \propto 1$ on $\mathcal{C}$.

(ii) C.2: current state $\in \mathcal{C}$; **C.4**: probability of generating $(\mathcal{B}, \mathcal{C})$ is the same for all $w \in \mathcal{C} \Rightarrow p(\mathcal{B}, \mathcal{C} \mid w, u, \varepsilon) \propto \mathbb{I}[w \in \mathcal{C}]$.

Substituting: $p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) \propto \mathbb{I}[w \in \mathcal{C}]$, i.e. uniform over $\mathcal{C}$.

By **C.1** (leapfrog is volume-preserving), all elements of $\mathcal{C}$ carry equal mass:

$$p(w \mid u, \mathcal{B}, \mathcal{C}, \varepsilon) = \frac{1}{|\mathcal{C}|}\, \mathbb{I}[w \in \mathcal{C}]. \qquad \square$$

Appendix: Benchmarks Bayesian & Hierarchical Logistic Regression Efficiency

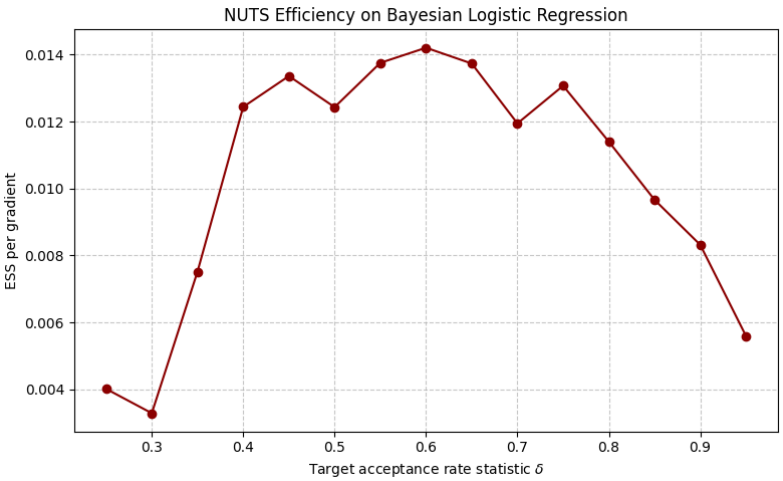


Figure 2: Bayesian Logistic Regression (25-D).

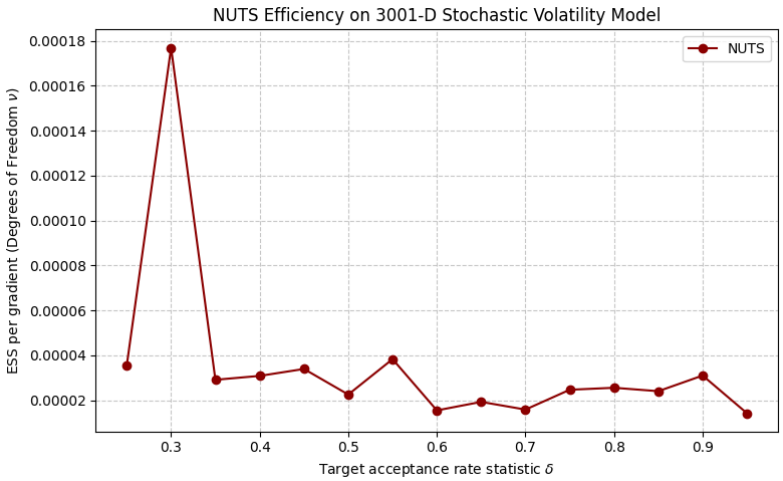


Figure 3: Hierarchical Logistic Regression (302-D).

Appendix: Dual Averaging Convergence

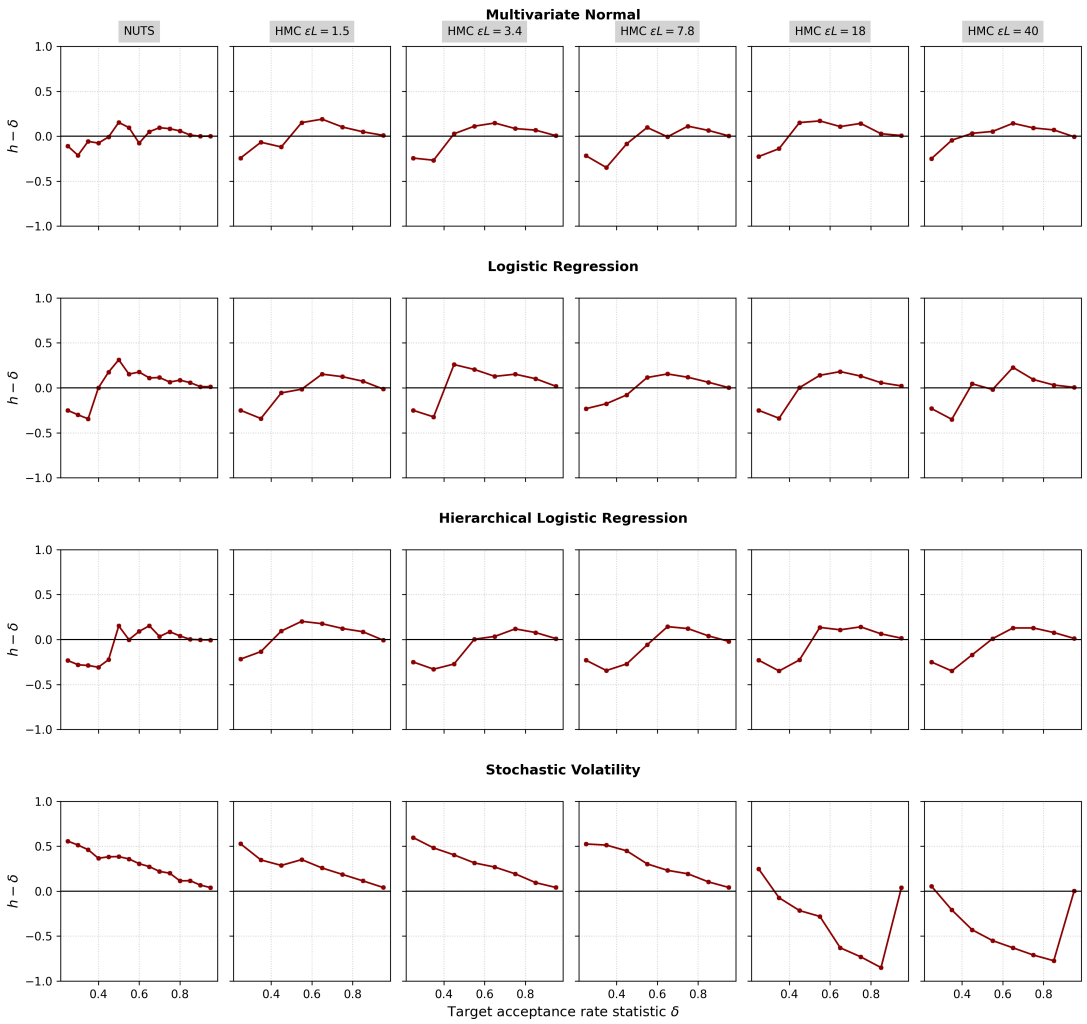


Figure 4: Discrepancy plots ($h - \delta$ across δ) for all four original benchmarks. Dual averaging reliably drives acceptance to target δ within a few hundred warm-up iterations.

Appendix: BNN Trace Plots

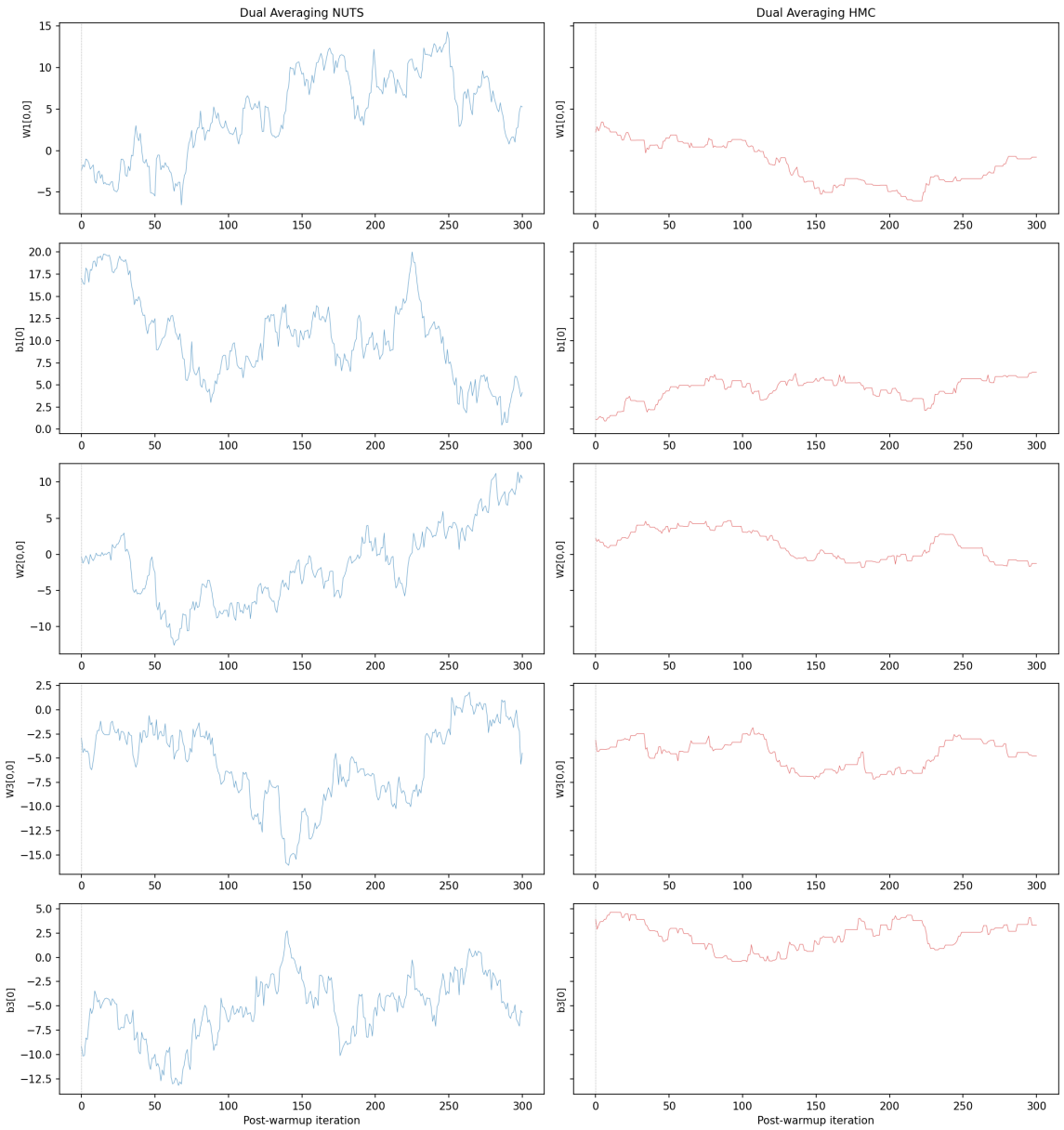


Figure 5: Trace plots: NUTS (blue) covers a broader range of weight values than HMC (red).

Appendix: BNN Adaptation Diagnostics

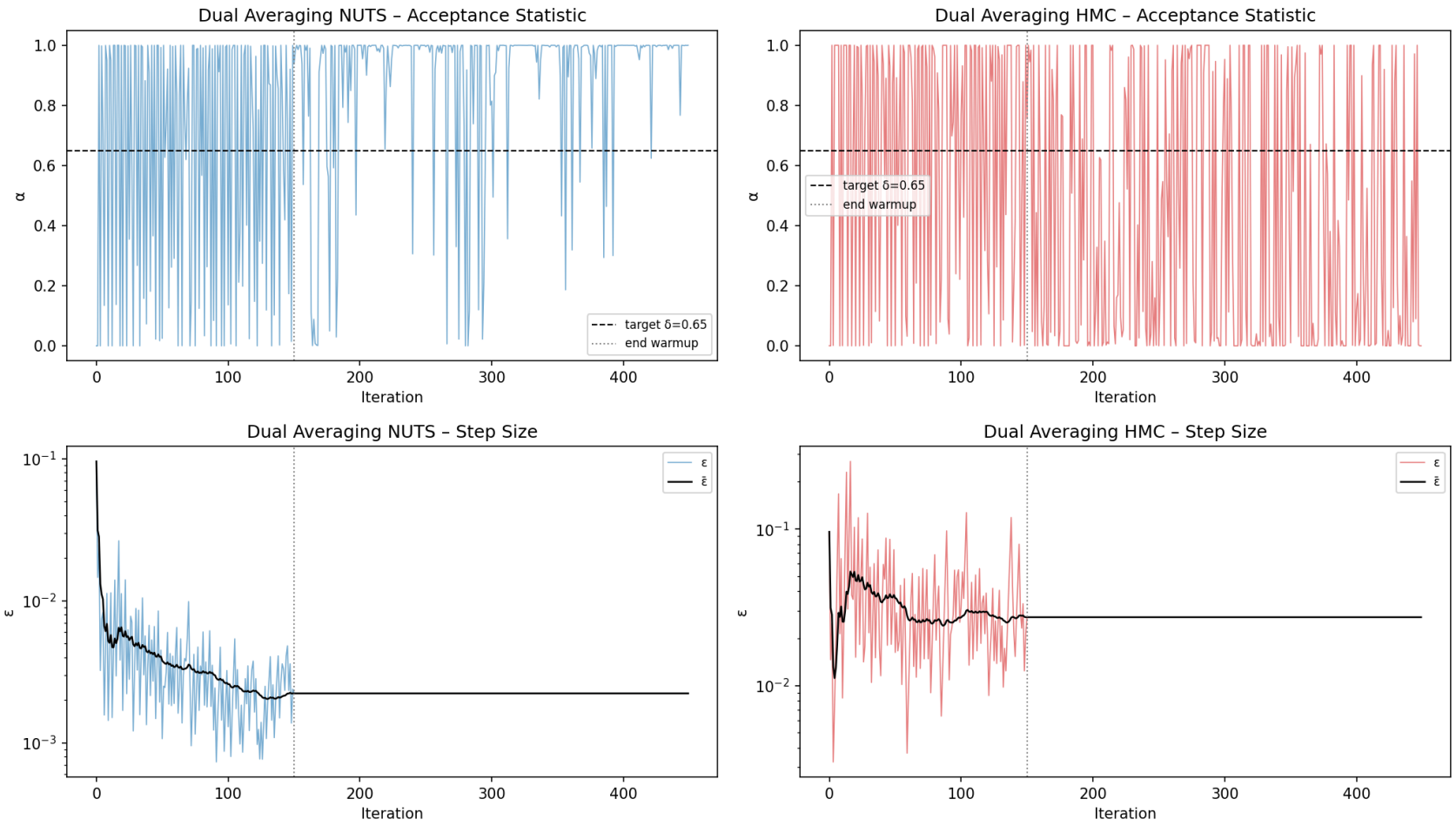


Figure 6: Dual-averaging convergence on the BNN. Step size $\bar{\varepsilon}$ stabilises within the warm-up phase.

BNN: Efficiency

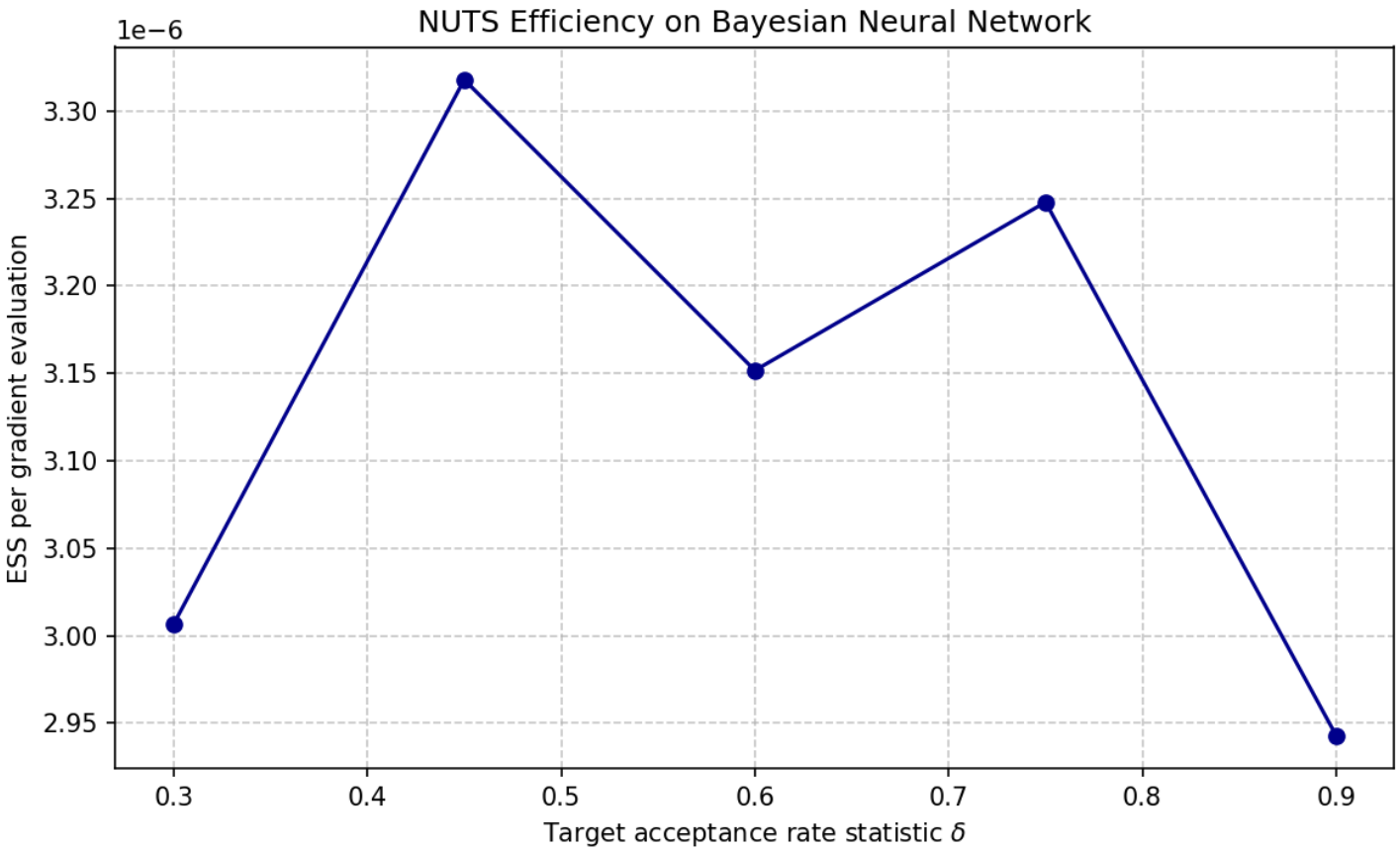


Figure 7: ESS/gradient on the BNN. NUTS consistently achieves higher effective samples per gradient evaluation than HMC across all tested configurations.

Appendix: GPC Kernel Matrix & Posterior Correlations

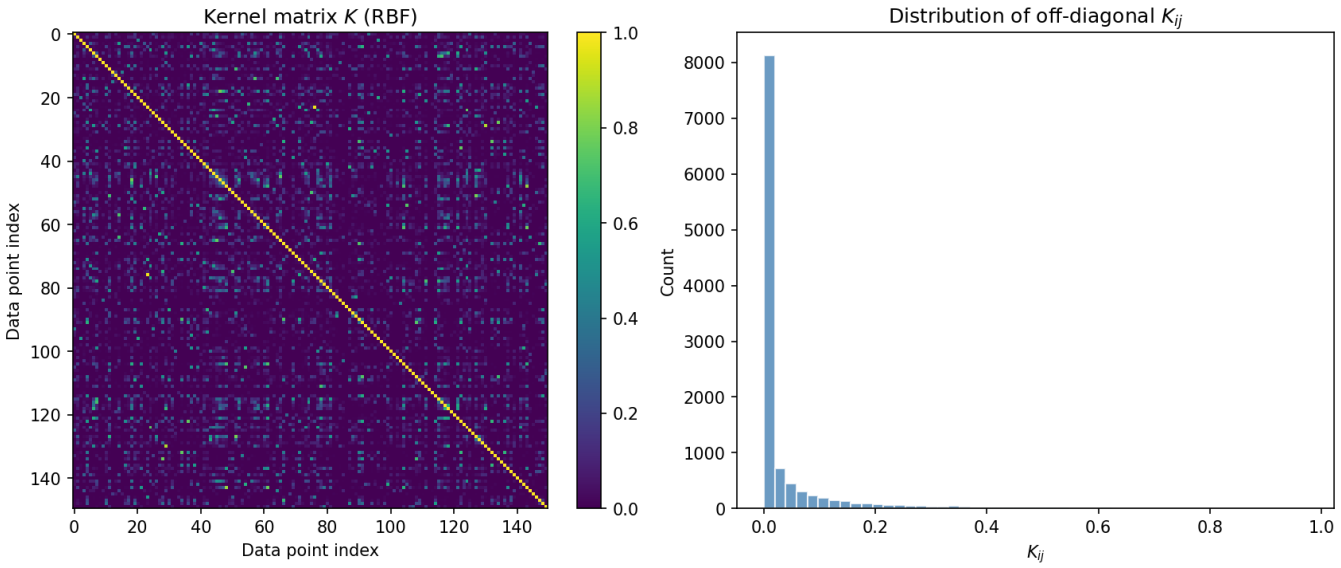


Figure 8: RBF kernel matrix K: dense correlation structure.

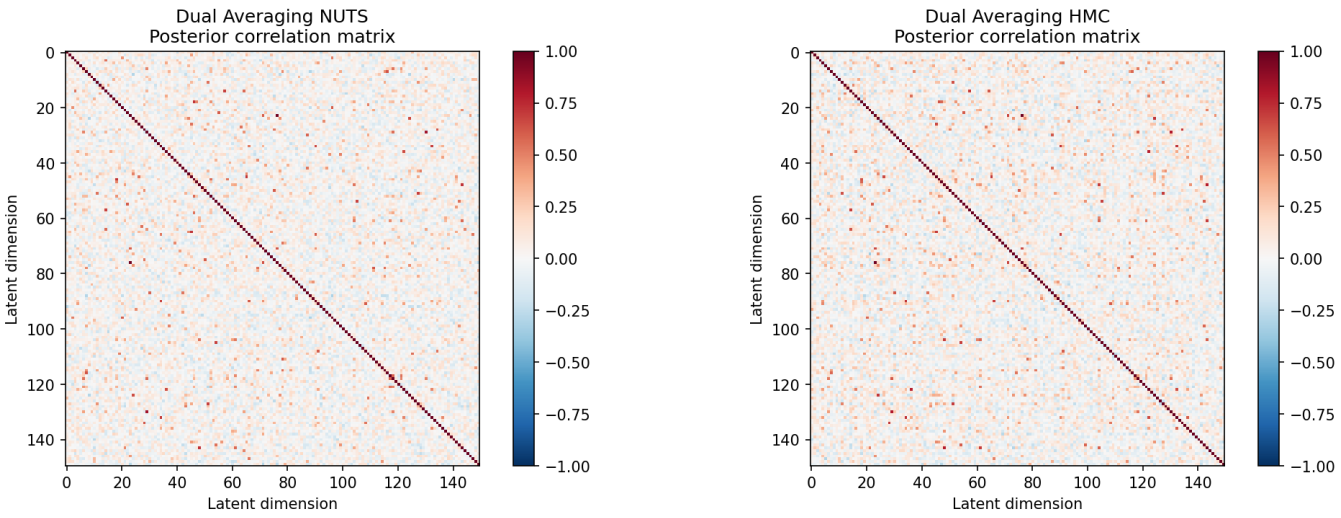


Figure 9: Posterior correlations among latent values f.

Appendix: GPC Efficiency

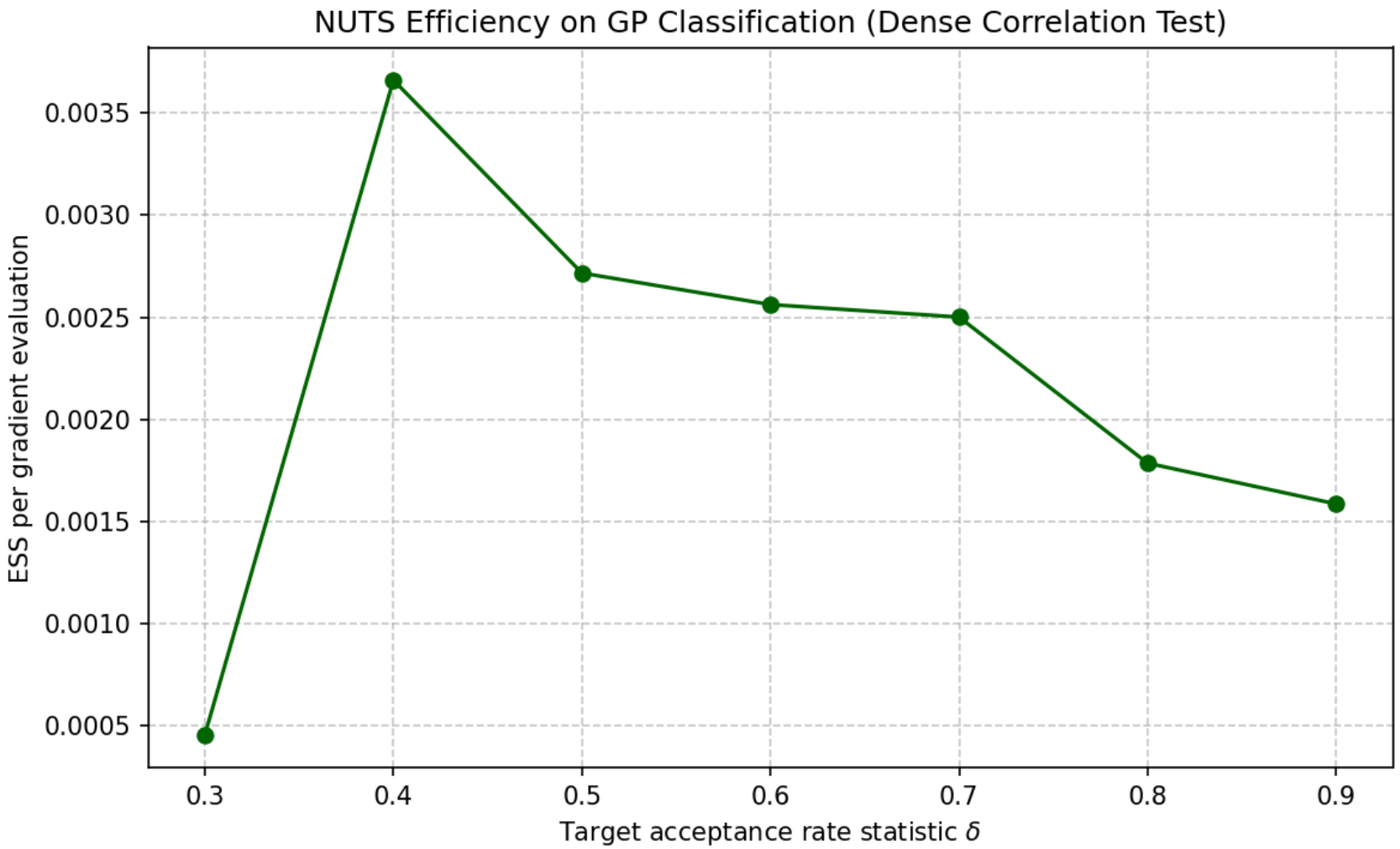


Figure 10: ESS/grad on GP classification. NUTS wins on per-gradient quality but at higher absolute cost.

Appendix: Funnel Scatter

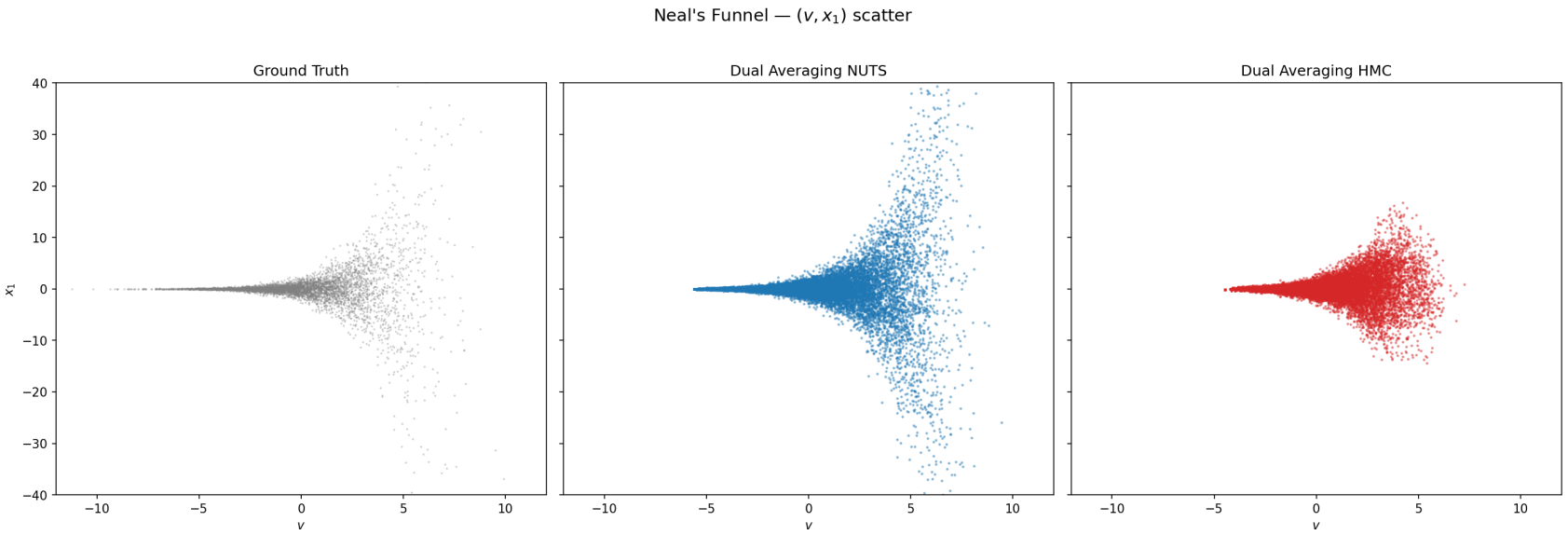


Figure 11: Left: ground truth. **Centre:** NUTS covers the full funnel. **Right:** HMC fails to explore it.

Appendix: Funnel Trace Plots & Marginal

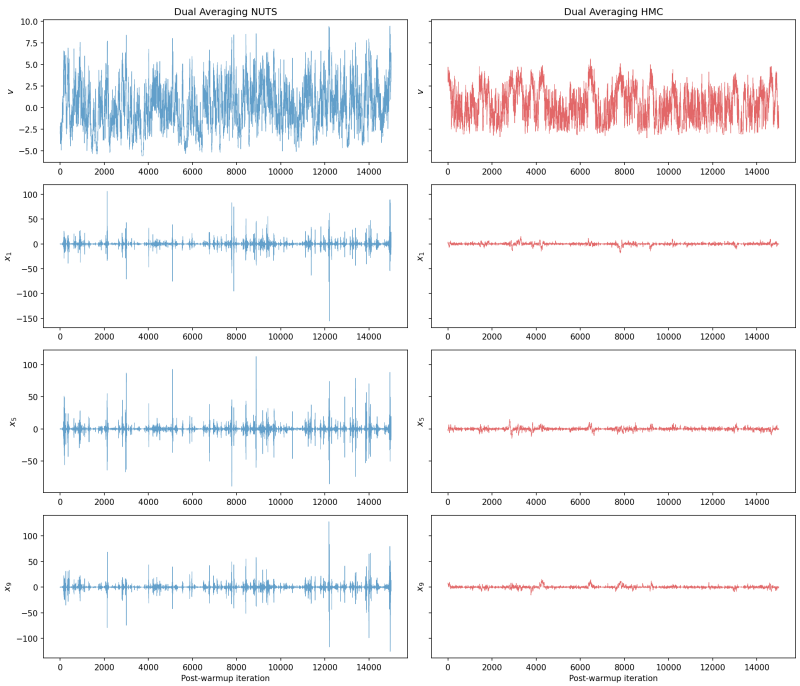


Figure 12: NUTS mixes freely; HMC shows flat segments.

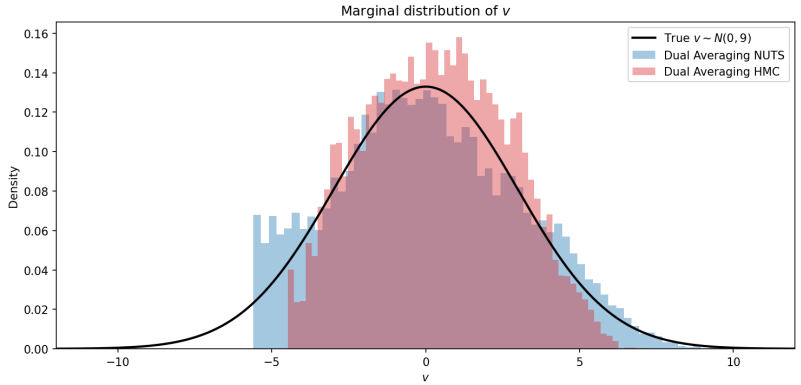


Figure 13: Marginal of v: NUTS recovers truth; HMC misses $v < 0$.

Appendix: Funnel Trajectory Lengths

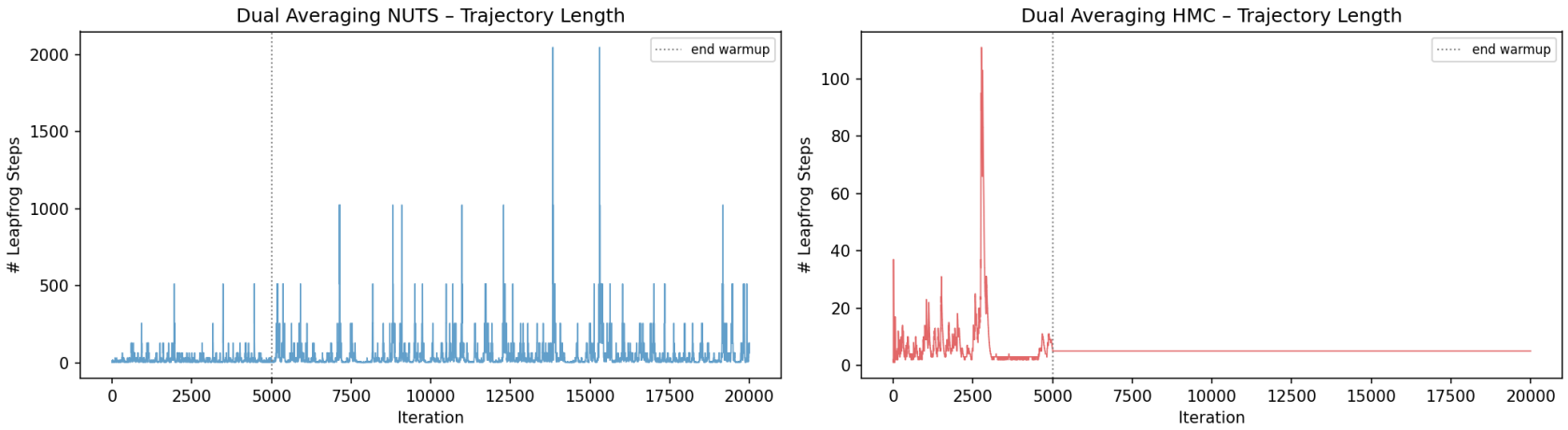


Figure 14: NUTS trajectory lengths on Neal's Funnel. Adaptive mechanism adjusts to local geometry.